Gizmonic Cambot: an open, $20 camera that remembers what your robots did

White paper · semantically indexed video for AI monitoring of robots and lab equipment

Gizmonic Cambot · open hardware and open source
Draft 0.2 · November 2026

Summary

Robots fail in ways their logs do not explain. A pipette misses a tip, a gripper drops a part, a mobile base clips a table leg. Video of the event usually exists only if someone happened to be recording, and even then no one wants to scrub through hours of footage to find the four seconds that matter.

Gizmonic Cambot is an open-source camera built from a roughly $12 ESP32-S3 camera board, a microSD card and a 3D-printed case. It records continuously into a rolling buffer on the card, uses cloud vision models on the device to build a searchable index of what happened, and answers questions in plain language from a LAN registrar on a nearby computer:

> “Show me between 12:00 and 12:30 where the machine picks up a pipette tip.”

The camera serves live frames, recordings, clips, its index and an MCP tool surface over HTTP. A small registrar on the LAN shows the live view, keeps a local copy of the index, and runs the ask loop so the camera stays free to record. The same camera tools are available to a coding agent such as Cursor through MCP.

The hardware, case, mounts, firmware and documentation are all open. The case uses a GoPro-style **W-Mount** so the camera can hang from modular arms ending in a dual-magnet foot, or any other W-Mount accessory.

The problem

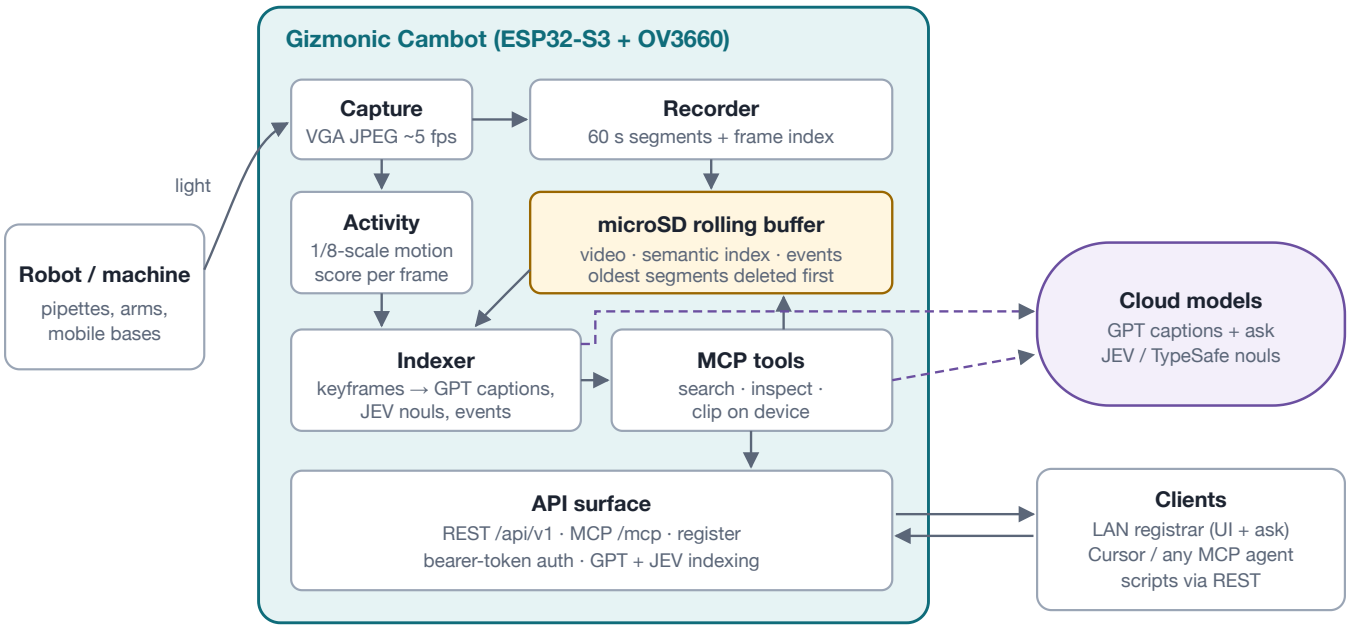
Labs and workshops are filling up with low-cost automation: liquid handlers, desktop arms, 3D printer farms, pick-and-place cells, hobby mobile robots. Three things make visual monitoring of them harder than it should be:

1. **Video is write-only.** Conventional cameras record well but cannot answer “when did X happen?” Finding an event means a human scrubbing a timeline.
2. **Cameras are closed or expensive.** Commercial “AI cameras” lock footage and models into a vendor cloud, cost $50–$300 each, and are designed for doorways rather than robot work cells.
3. **Mounting is an afterthought.** A camera that watches a robot needs to sit on the robot’s frame, a shelf edge, an aluminum extrusion or a steel enclosure, often at an odd angle. Most cameras ship with one tripod thread.

The concept

Cambot is designed to be cheap enough to leave on every machine:

- **Records everything locally.** About five VGA JPEGs per second by default (640×480, 200 ms pacing), stored as one-minute segments with a per-frame time index. When the card fills, the oldest hour of video is deleted first. Nothing leaves the camera unless indexing or a query sends it.
- **Indexes only what matters.** An on-board activity detector scores motion on every frame. Keyframes go to a vision model when something is happening, plus one idle keyframe a minute, so an empty room costs almost nothing.
- **Two index paths.** GPT captioning writes structured captions, objects and actions. JEV (TypeSafe) scores named **nouls**—yes/no watch questions with optional example images, bounding boxes and webhook alerts. Either path can run alone.
- **Answers questions off-camera.** The LAN registrar syncs index history into SQLite and asks gpt-4.1-mini against that copy, so a long question does not stall recording. The camera still exposes MCP tools for agents that want to search and inspect frames themselves.
- **Speaks to agents.** REST `/api/v1/...` and MCP `/mcp` share the same capabilities: status, recordings, index search, clips and live frames.



System architecture. Capture, record and index run on the ESP32-S3; ask and the live UI run on the LAN registrar.

How it works

Capture and the rolling buffer

The OV3660 sensor produces JPEG frames directly, so the ESP32-S3 never encodes video. The recorder appends frames to a one-minute segment file and writes a small index of each frame's timestamp, offset, length and motion score. Timestamps come from network time (SNTP) and are stored in UTC. Default capture is **VGA at ~5 fps**. A **High performance** flag on the card (and in the registrar UI) keeps that rate while flushing the card less often and refreshing the live preview every 0.5 s.

Mode	Resolution	Rate	Notes
Default	640×480 (VGA)	~5 fps	Chosen so TypeSafe image sides stay ≥ 280 px
High performance	640×480	~5 fps	Faster preview; lighter SD flush

Higher resolutions and rate tables from the original design brief remain aspirational. Busy, detailed scenes produce larger JPEGs and fill the card faster.

The semantic index

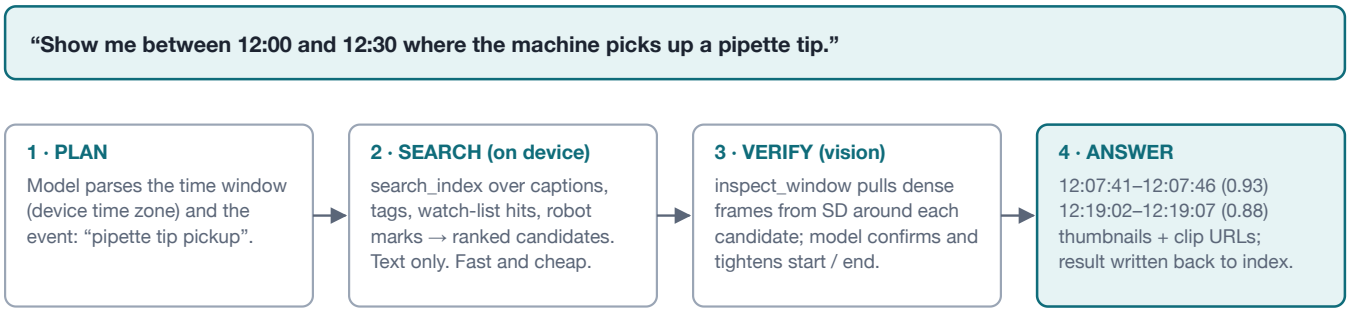
Each keyframe the on-device indexer selects can be sent to:

- **GPT** (`gpt-4.1-mini` today) with a short scene profile, returning caption, objects, actions and robot state.
- **JEV** with the camera's noul list. Each noul is a yes/no question; optional example JPEGs, a box question and an HTTPS alert webhook travel with it.

Records land under `/sdcard/cambot/index/` as hourly JSONL. Consecutive keyframes that share an action merge into **events**. Nouls live in `nouls.json` (plus example JPEGs under `nouls/<name>/`). GPT and JEV indexing are separate switches (`/api/v1/indexing`, `/api/v1/jev`).

Robot marks (`POST /api/v1/marks`) remain planned. Controllers will be able to stamp the timeline with labels such as `step 12: pick tip` so the camera can answer those queries exactly.

Answering a question



How a question becomes an answer. Only frame inspection sends images to the model.

Today the ask path runs on the **registrar**:

1. **Sync.** The registrar pulls recent index lines from the camera (including JEV watch scores) into a local SQLite history so questions can cover days without hammering the ESP32.
2. **Search.** gpt-4.1-mini reads that history for the time range in the question.
3. **Inspect (optional).** Agents using MCP on the camera can still call `inspect_window` / `make_clip` to pull frames or an MJPEG clip from the card when the text index is not enough.
4. **Answer.** The registrar returns moments with the supporting index text. Clip URLs point at the camera.

An on-board `/ask` agent that owns the full tool-calling loop on the ESP32 is still on the roadmap. Until then, MCP on the camera and ask on the registrar cover the same jobs for people and for coding agents.

Interfaces

Interface	Used by	What it offers today
Registrar at `http://<host>:8765/`	People	Live view, caption/noul UI, GPT/JEV/high-perf switches, ask box, index history
REST `/api/v1/...`	Scripts, registrar	Status, frame, recordings, clip.mjpg, index/latest, indexing, jev, nouls, perf
MCP `/mcp`	Cursor and other agents	`get_status`, `list_recordings`, `search_index`, `read_index_page`, `inspect_window`, `make_clip`, `get_live_frame`
`cambot` CLI	Setup and bulk work	**Not shipped yet** (milestone M2). Use ESP-IDF + `fixr/SETUP.md`

Example uses

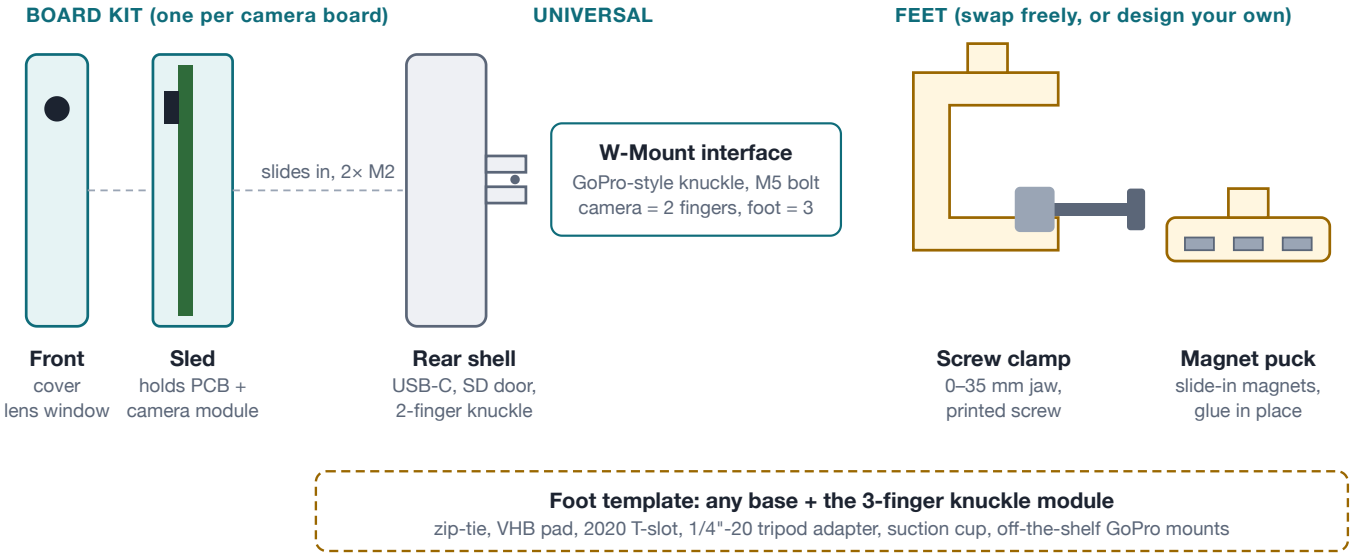
- **Lab automation.** “List every tip pickup this morning that took more than one attempt.” “Did anyone open the enclosure during the 2 PM run?”
- **Robot development.** Mount the camera over a work cell and ask what the robot did during a run, without scrubbing through raw video by hand.
- **3D printer farms.** Nouls for “spaghetti”, “part detached” and “nozzle blob” turn the index into an alert feed via webhook.
- **Pick-and-place and QA cells.** Count picks per hour, pull clips of dropped parts, and (once marks land) compare robot claims to what was seen.
- **Agent-driven debugging.** A coding agent fixing robot firmware can ask the registrar what the camera saw after the last flash.

Hardware

The reference board is the **FORIOT ESP32-S3-CAM** (Amazon B0GD2BZ9B1), a common layout also sold by Freenove, Goouuu and Keyestudio:

Item	Specification
Module	ESP32-S3-WROOM-1 N16R8: dual-core 240 MHz, 16 MB flash, 8 MB octal PSRAM
Camera	OV3660, 3 MP, on a 24-pin DVP flex cable
Storage	microSD (push-push, rear), 1-bit SDMMC on GPIO 38/39/40
Connectivity	2.4 GHz Wi-Fi 802.11 b/g/n, Bluetooth LE 5
USB	Two USB-C: TTL (CH340 UART, for flashing and logs) and native USB OTG
Indicators	Power LED, WS2812 RGB LED (GPIO 48), user LED (GPIO 2)
Size	62.6 × 28.3 mm PCB (vendor figure)
Power	5 V USB, plan for 0.5 A

Case and modular mounts



The case is split so only two small parts change per board. Everything that touches the outside world is universal.

The printable kit that ships in this repository (`hardware/cambot-minimum-petg-kit/`, release 2026-11-04) is a **minimum PETG mount**, not the earlier sled / front-cover / clamp design:

- **housing_v3** — open drop-in body with two center posts and a W-Mount knuckle; no front cover.
- **retainer_v3_short** — snaps over the PCB to hold it on the posts.
- **Modular arms** — male/female rods (25 mm and 50 mm shafts) locked with a 6-32 set screw.
- **dual_magnet_foot** — three-finger W-Mount base with two external rubber-coated magnets under the plate.

Print the three plate 3MFs, assemble with M5 pivots and M4 magnet fasteners, and stick the foot to steel. Off-the-shelf GoPro mounts also mate with the W-Mount. Details, PETG settings and hardware are in the kit README. The mechanical specification tracks that kit as the current design.

Cost

Part	Approx. cost
ESP32-S3-CAM board with OV3660 (from a 3-pack at $34.99)	$11.70
32 GB high-endurance microSD	$6–9
PETG for housing, retainer, arms and foot	$1–2
M5 bolts/nuts, 6-32 screw, M4 magnet hardware, 2 rubber-coated magnets	$5–10
USB-C cable and 5 V supply (often already on hand)	$0–6

Part	Approx. cost
Total	**about $25–40**

Cloud cost depends on activity, not on recording time. Each captioned keyframe or JEV call costs roughly as much as one small image prompt. The firmware enforces a daily frame budget; recording continues when indexing pauses.

Privacy and security

- Video stays on the SD card. Images go only to the model endpoints the owner configures (OpenAI-compatible and/or TypeSafe), and only when indexing or a query sends them. Captioning and JEV can each be switched off.
- The model endpoint can be any OpenAI-compatible server, including a model running on the local network, so GPT images need not leave the building.
- The API requires a bearer token (the camera password). The camera makes outbound connections to Wi-Fi, SNTP, the registrar and the model hosts. It is meant for a trusted LAN, not for exposure to the internet.
- Today secrets are compiled into `firmware/main/secrets.h` for bring-up (copy from `secrets.h.example`: `CAMBOT_WIFI_*`, `CAMBOT_CAMERA_PASSWORD`, `CAMBOT_REGISTRAR_URL`, `CAMBOT_TZ`, optional `CAMBOT_MODEL_KEY` / `CAMBOT_JEV_KEY`). NVS-backed provisioning and encrypted builds are still planned.

Limitations

- **2.4 GHz Wi-Fi only**, WPA2 or WPA3 Personal. No enterprise (802.1X) or captive-portal networks in version 1.
- **Frame rate.** ~5 fps suits robot motion at human speeds. Very fast events can fall between frames; robot marks are the planned fix.
- **Sparse indexing.** The index is only as good as its keyframes and nouls. MCP inspect/clip cover the gaps at extra cost.
- **No audio and no H.264.** The ESP32-S3 has no video encoder; clips are MJPEG.
- **Clock.** The board has no battery-backed clock. Until network time is synced after boot, segments are marked unsynced.
- **Ask lives on the registrar** for now, not as `POST /api/v1/ask` on the camera. The `cambot` CLI, SoftAP provisioning, on-camera web UI and OTA are not shipped yet.
- **Model accuracy.** Captions and noul scores can be wrong. Every answer links to the footage so a person can check it.

Open source plan and roadmap

Item	License (proposed)
Firmware, host libraries, registrar	Apache-2.0
Case, mounts, CAD source and STL	CERN-OHL-P-2.0
Documentation	CC BY 4.0

Milestone	Status
M0–M1 Board bring-up, SD rolling buffer, REST, live frame, clips, eviction	**Done** in `firmware/`
M2 Wi-Fi provisioning, `cambot` CLI, browser flasher	Not started; `fixr/SETUP.md` + secrets.h is the path today
M3 Activity, on-device GPT + JEV index, nouls, registrar live UI	**Done**
M4a MCP on camera; registrar ask + index history	**Done**
M4b On-board `/ask` agent, marks, full index search API	Planned
M5 OTA with rollback, encrypted builds, second board kit	Planned
Later Embeddings, multi-camera hub, clamp foot, front cover	Planned

Companion documents: the **mechanical specification** (current PETG kit), the **firmware repository plan** (tree, API and remaining work), and the **setup guide** in `fixr/SETUP.md`, written so a person or a coding agent can follow it.